

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/MANAGEMENT
/COMMERCIAL PRACTICE**

**INTRODUCTION TO HYBRID AND ELECTRIC VEHICLE ENGINEERING
MODEL QUESTION PAPER**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List any two advantages of electric vehicles	M1.01	R
2	Which of the following vehicle is most efficient? a. Hybrid Electric Vehicle b. Fuel vehicle c. Electric Vehicle	M1.04	R
3	The electric motor in hybrid electric vehicle is also acts as a _____.	M2.03	R
4	List any two methods by which batteries of plug-in hybrid vehicles are charged.	M3.01	R
5	What is the lifespan of electric car batteries? a. 10-20 years b. 20-30 years c. 5-10 years d. 50 years	M3.01	R
6	 Select the CORRECT statement for the image shown. This is a _____. a. Inverter and converter unit b. High voltage cable c. High voltage battery d. Motor Generator	M3.04	U
7	Electric Vehicles are generally powered by _____ batteries.	M3.01	R
8	Expand CAN.	M4.01	R
9	List the three types of EV charging	M4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module outcome	Cognitive level
1	Give Comparisons of Hybrid Electric Vehicles and Conventional Vehicles?	M1.01	U
2	Explain different types of forces acting on vehicle going uphill.	M1.02	U
3	List economic and environmental impact of electric hybrid vehicle.	M1.04	R
4	With the help of block diagram explain the major components of an electric vehicle	M1.03	U
5	What are the desired features of motors used for electric vehicles.	M2.02	U
6	Explain the terms specific energy and energy density applied to batteries.	M3.01	U
7	What are the factors affecting the performance of batteries used in EVs?	M3.01	U
8	Explain the significance of deep cycles and depth of discharge of batteries associated with in EV/EHVs.	M3.01	U
9	Explain the working of super capacitors. Give its advantages and disadvantages.	M3.02	U
10	What is the significance of a communication network in electric/hybrid vehicles?	M4.01	U

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

		Module outcome	Cognitive level
1.	Differentiate between complex hybrid and series parallel hybrid configurations	M1.03	U
	OR		
2.	What are the social and economical impacts of hybrid vehicles? Under what condition a pure EV can be chosen as a better option compared to hybrid vehicles considering the impact of climate change?	M1.04	U
3.	Draw six different configurations of drive trains in electric vehicles. Briefly explain each configuration.	M2.01	U
	OR		
4.	With the help of neat figures explain the general configuration of constant v/f control of induction motors.	M2.03	U
5.	Draw the block diagram of electric propulsion unit of electric vehicle and mention the relevance of each block.	M2.02	U
	OR		
6.	Discuss different methods of control used for DC-DC controlled drives in EV/HEV.	M2.03	U

7.	What type of operation is known as flux weakening in permanent magnet machines? List the major advantages of permanent magnet machines in EV.	M2.03	U
OR			
8.	Discuss different methods of control used for DC-DC controlled drivers in EV/EHV.	M2.04	U
9.	Explain the operation, advantages and disadvantages of fuel cells.	M3.02	U
OR			
10.	What is meant by C-rating of a battery? If a 100Ah battery is rated C_5 what would be its discharge current expressed as $0.5C_5$?	M3.01	U
11.	Compare different energy management strategies used in EV/EHV.	M4.02	U
OR			
12.	Why an energy management control system is required in EHV? Do you think an elaborate energy management system similar to that applied to a hybrid vehicle is required in electric vehicle. Explain.	M4.02	U